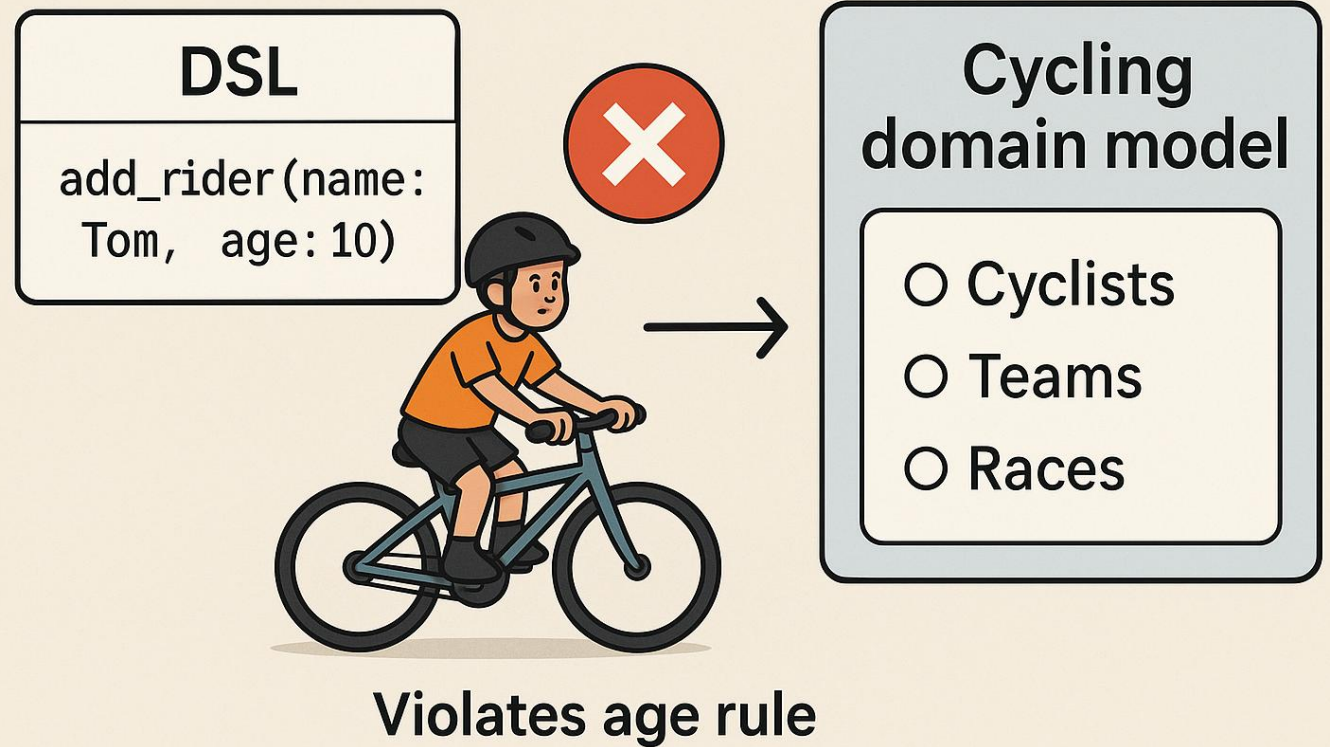


MAKING DSLS CONVERSABLE

Krishna Narasimhan,  F1RE 
ignite change

DSLs are great

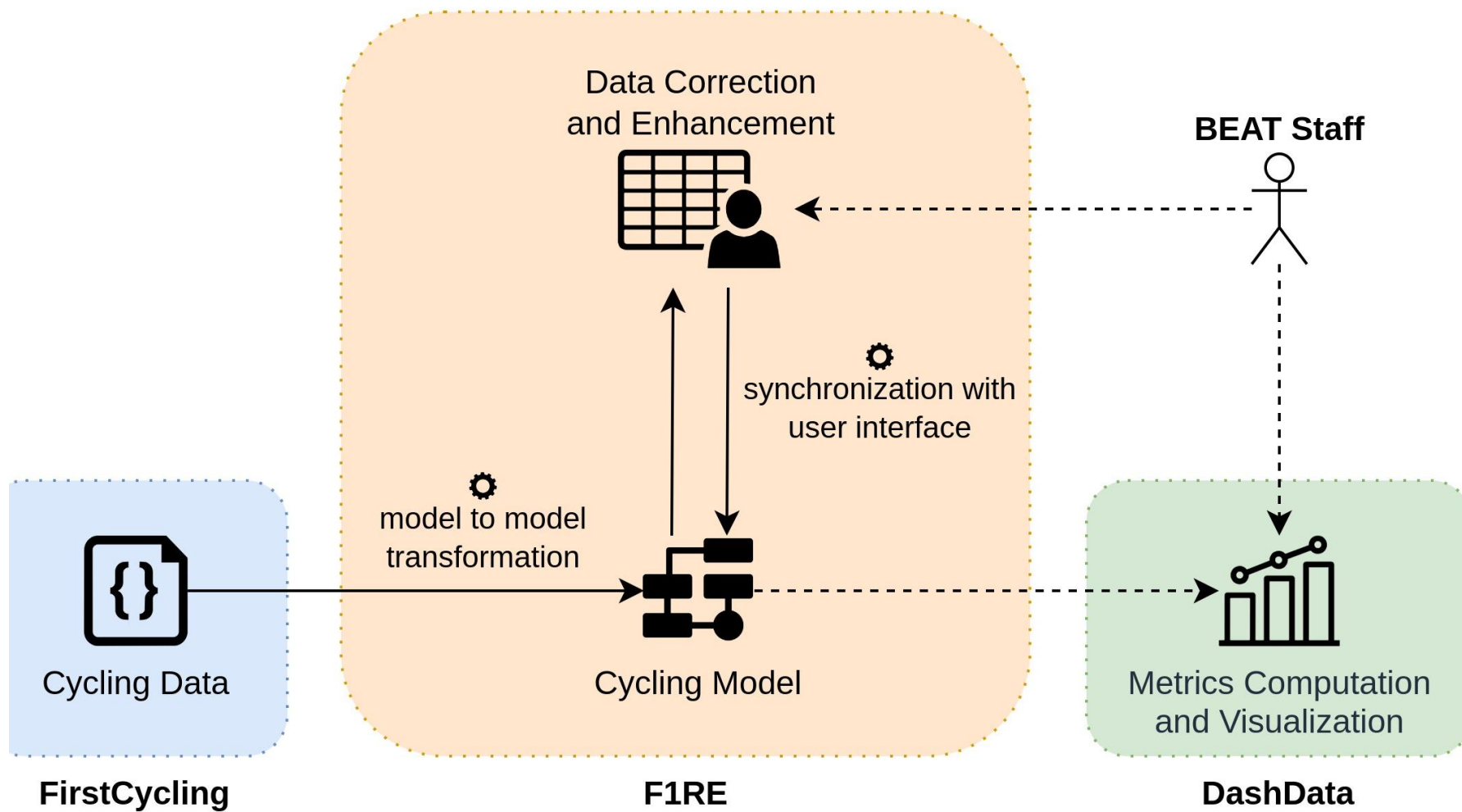
- DSLs model complex domains with precision.
- They ensure valid, rule-consistent operations.
- Example: cycling domain model enforcing age limits or team structure.



Use case: BEAT Cycling Club

- Talent prediction for cycling pro's
- Race results combined with secondary data (weather, location, training data, equipment)
- Problem:
 - Users without ANY tech skills
 - Users have very limited time

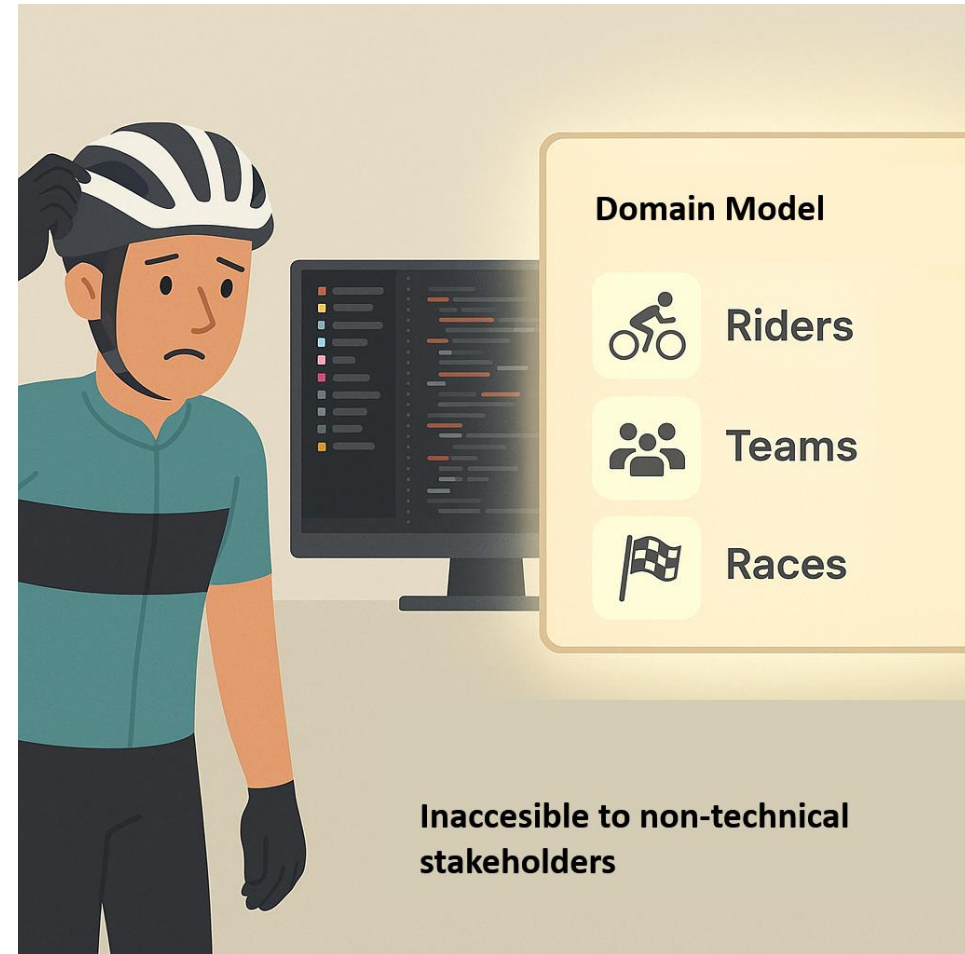




Accessibility issue

- Complex syntax
- Special tooling required
- Limited documentation
- Steep learning curve

Risk of expensive shelf-ware



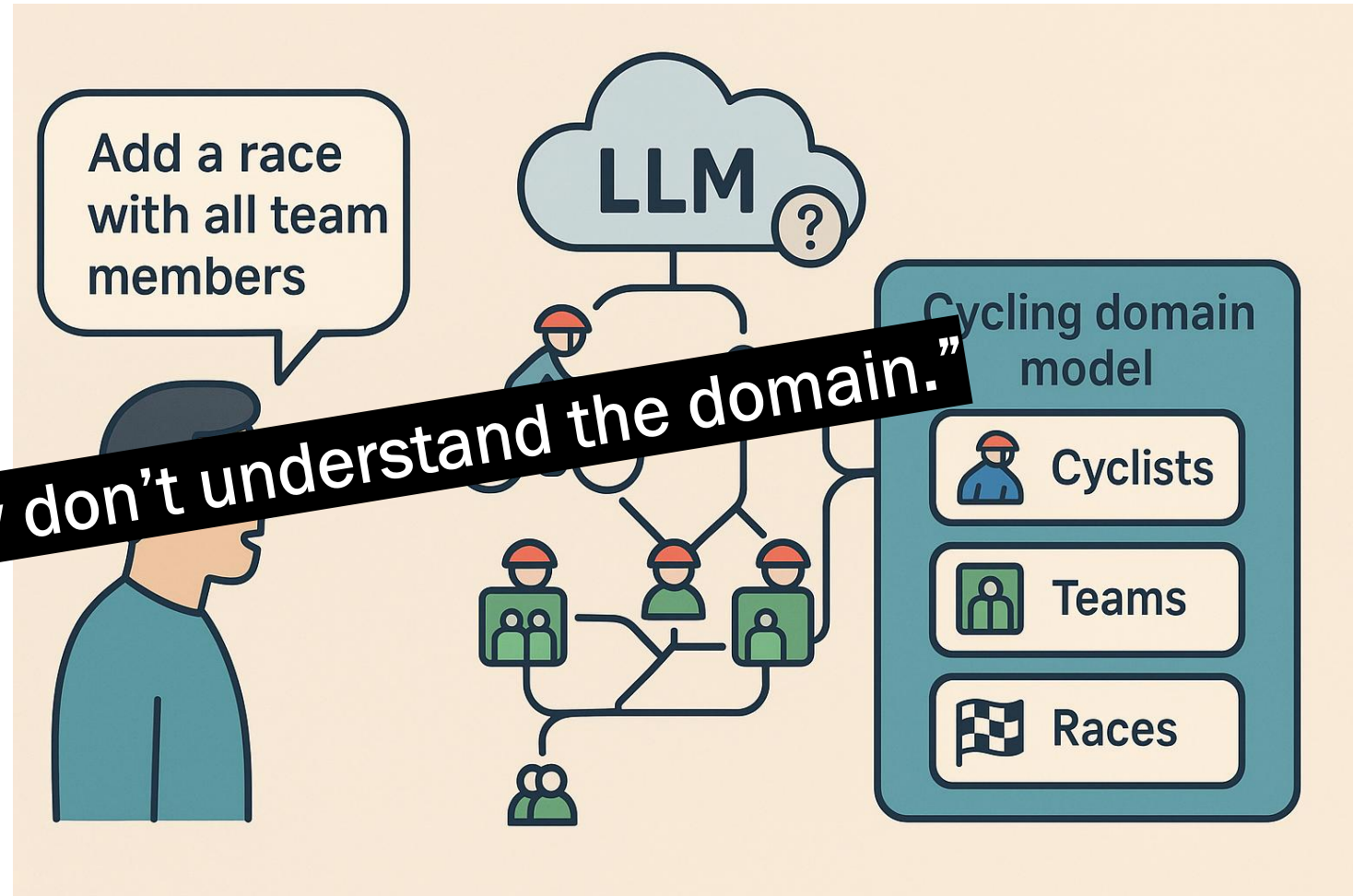
Why not teach* an LLM and ask it for help?

** Fine-tuning, Prompt prefix, RAG*

LLMs ≠ Domain Understanding

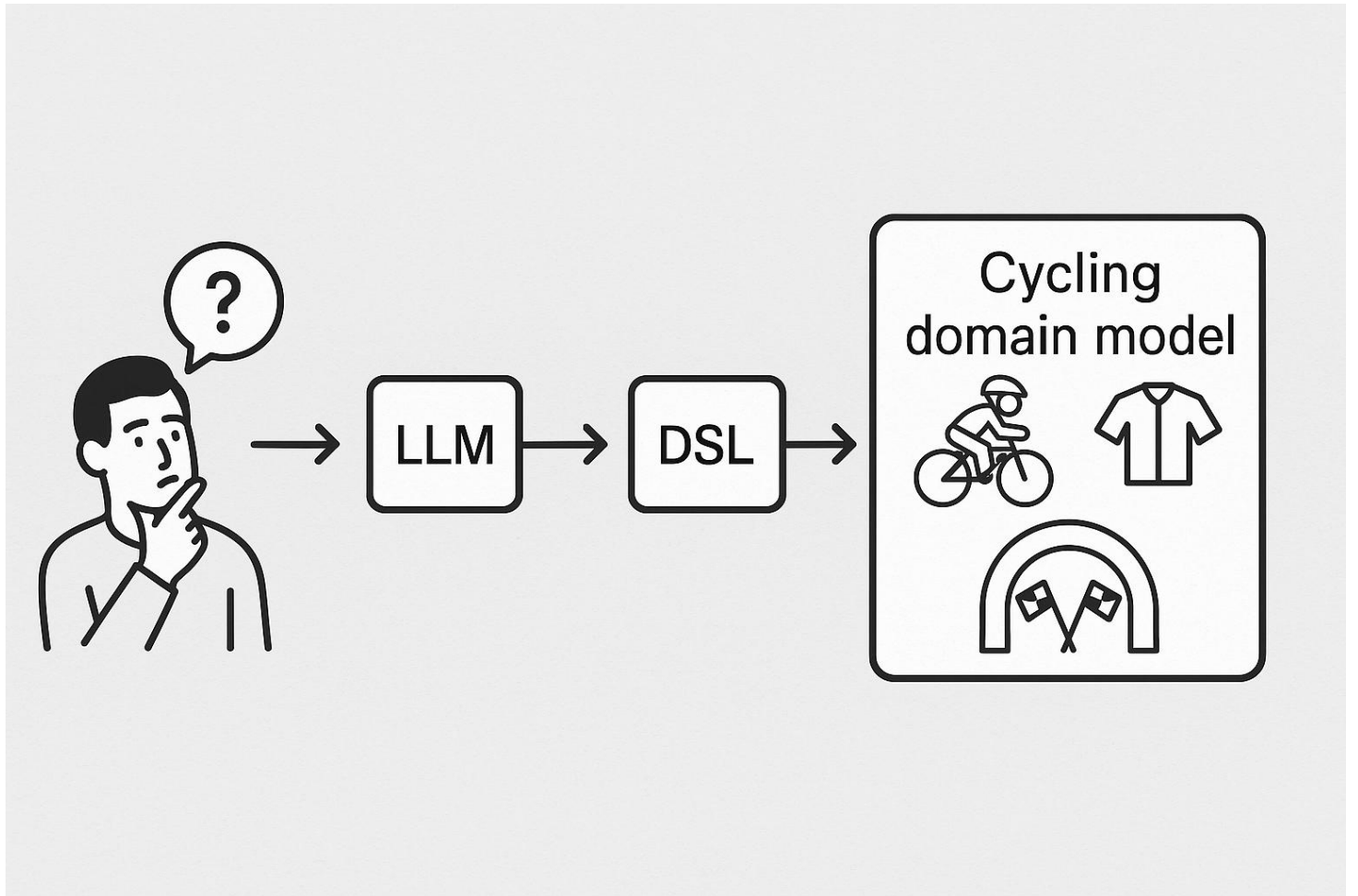
- Words, not structure
- Confuses relationships
- No rule awareness
- Sounds right, acts wrong

“LLMs can talk, but they don’t understand the domain.”



Best of both worlds – Separation of concerns

- LLMs offer accessibility — natural, flexible, easy for anyone to use
- DSLs offer reliability — structured, validated, rule-bound
- What if each did what it's best at?
- LLMs understand intent → DSLs ensure correct execution
- Together: Conversational DSLs — natural input with guaranteed validity



Demo (first part - to show how the envisioned system could look like)

- Making the point “If we could use natural language to interact with the underlying model, this is what it would look like”.
 - Show how an input like this results in model change
- ““Let's set up the cycling team for next season. Create a rider named Erkan Dikan from Turkey aged 23. Create another rider Priya from Germany. Also create Team SkyIN from India with a budget of 20 million founded in 2010.”

Assistant: Found 3 task(s) in the document. I'll process them one by one.

Processing Task 1/3: Create rider Erkan Dikan

Here is what I understood:

Role: admin

Action: Create Rider

Available Input:

- Name: Erkan Dikan
- Age: 23
- Country: Turkey

Review and confirm details for 'Create Rider':

Name (Extracted - edit if incorrect):

What is the cyclist's name?

Erkan Dikan

Age (Extracted - edit if incorrect):

How old is the cyclist?

23

Country (Extracted - edit if incorrect):

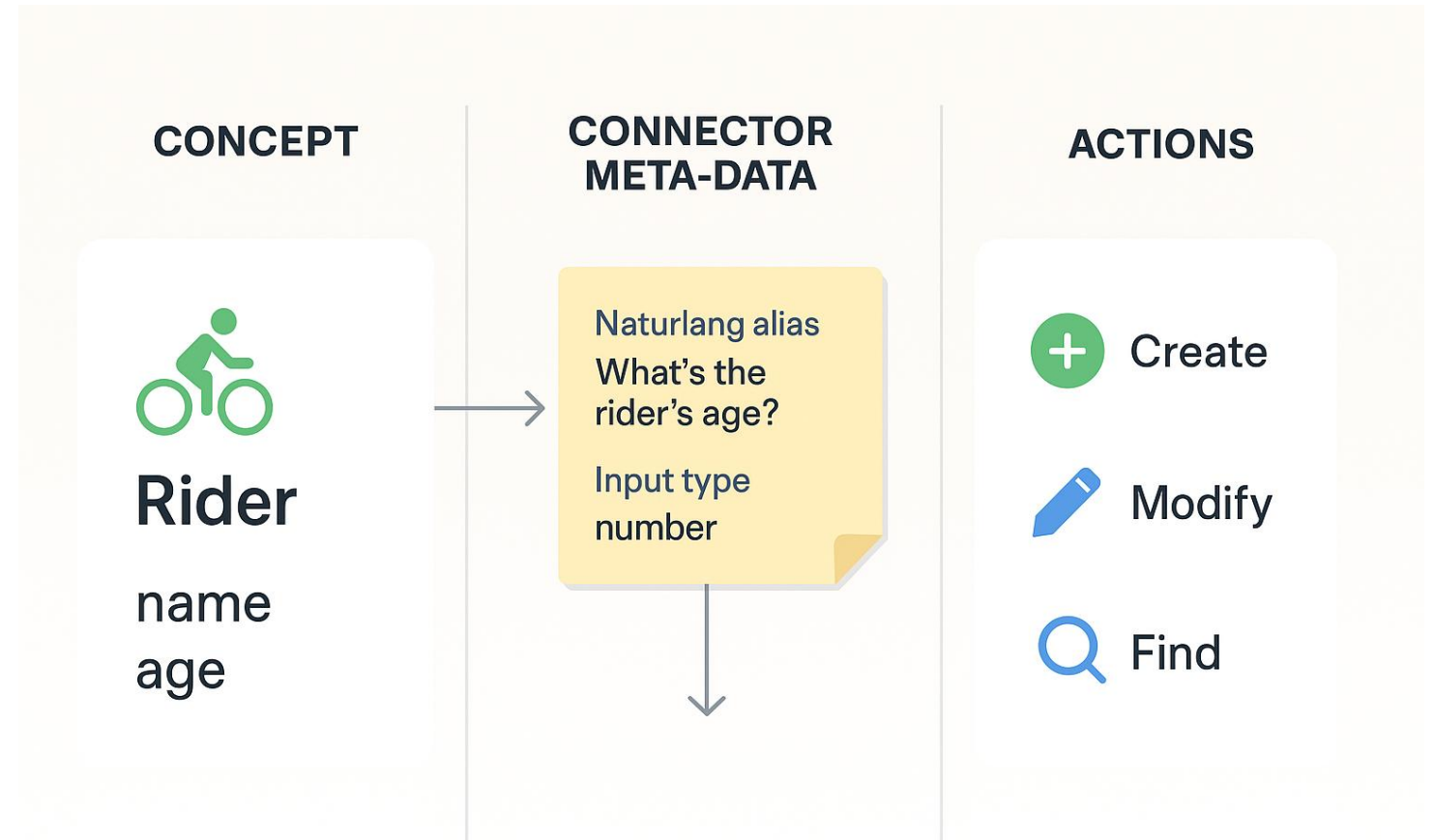
Which country is the cyclist from?

Turkey

✓ Submit and Ex...

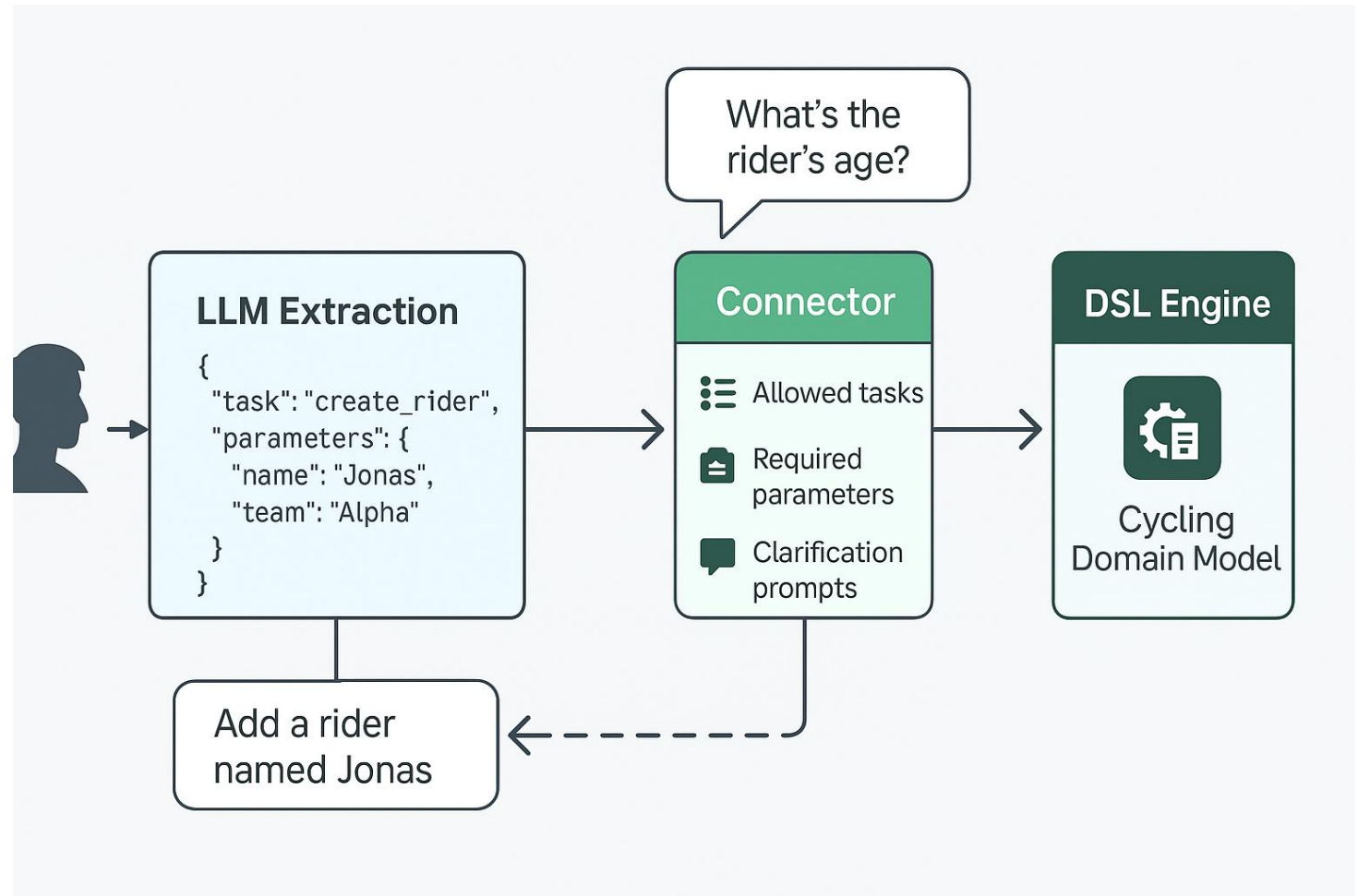
Keeping roles separate with Connector

- LLM never touches domain
- DSL defines all rules
- Connector bridges the gap.



High level system design

- LLM extracts intent
- Connector validates
- Missing info → clarification
- DSL executes valid actions
- Domain model = source of truth



Implementation

- Two reference implementations
 - LionWeb (<https://lionweb.io/>)
 - Lark (<https://lionweb.io/>)
- Supported model hubs:
 - Hugging Face
 - Ollama Cloud
 - Ollama locally hosted





Demo part 2

- Show multiple queries (create, modify, find) in both cycling (lionweb) and event management (lark)